Allouche Mohamed Abderrahmane

Software Engineer — Cybersecurity & Application Security

Kraków, Poland | +48 504 916 300 | abderrahmaneallouchee@gmail.com
allouche.dev | github | linkedin

Professional Summary

Software Engineer with 3+ years of experience developing and securing scalable web applications, with core focus in Application Security (AppSec), secure REST API development, and vulnerability mitigation. Proficient in OWASP Top 10 remediation, secure code reviews, SIEM log analysis with Wazuh, and penetration testing tools (Burp Suite, Nmap, Wireshark, Kali Linux). Experienced in hardening backend architectures in Python, FastAPI, React, PostgreSQL, and Docker. Ranked 3rd in Computer Science cohort of 120+ graduates; currently pursuing a B.Sc. in Cybersecurity at WSB University in Kraków.

Cybersecurity & Technical Skills

Security Domains: Application Security (AppSec), OWASP Top 10, MITRE ATT&CK, NIST CSF, Vulnerability Assessment, Threat Detection, SIEM Log Analysis, Secure Code Review, IAM
Security Tools: Wazuh (SIEM), Burp Suite, Nmap, Wireshark, Metasploit, Kali Linux, OWASP ZAP
Authentication & Cryptography: JWT, OAuth 2.0, Role-Based Access Control (RBAC), Argon2 Password Hashing, SSL/TLS, Public-Key Cryptography
Networking & Protocols: TCP/IP, DNS, HTTP/HTTPS, WebSockets, Firewalls, Routing & Switching, CCNA Network Security
DevSecOps & Systems: Linux/Unix CLI, Bash Scripting, Docker Container Security, Git, CI/CD Pipelines, Secret Management
Programming & Databases: Python (Security automation), SQL (SQLi remediation), TypeScript, JavaScript, FastAPI, PostgreSQL

Work Experience

SocShield DZ — Hybrid, Algeria
Full-Stack Developer — Application Security — Aug 2023 – Present

- Enforced OWASP Top 10 security standards across a cybersecurity SaaS platform, identifying and mitigating critical vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), and Broken Access Control.
- Architected multi-tier authentication and Role-Based Access Control (RBAC) across 4 user privilege tiers, enforcing strict authorization boundaries across administrative endpoints.
- Deployed and configured Wazuh SIEM agents to aggregate security logs, detect anomalous authentication spikes, and enforce file integrity monitoring across server infrastructure.
- Conducted vulnerability assessments and manual security scans using Burp Suite and Nmap, authoring technical remediation reports for development teams.
- Performed secure code reviews across 20+ backend REST API endpoints, implementing parameterized SQL queries and strict Pydantic validation schemas to eliminate injection flaws.

FlyRank Corp. (FlyRank AI) — Remote, USA
Back-End AI Engineering Intern — July 2026 – Aug 2026

- Engineered backend RESTful API endpoints in Python and FastAPI during a 6-week intensive program, implementing strict request validation and secure error handling.
- Audited asynchronous data pipelines and API workflows, enforcing least-privilege access and environment variable isolation for confidential model credentials.
- Containerized backend microservices with Docker, configuring non-root user execution and minimal base images to reduce container attack surfaces.

Security Projects

LeakShield – OSINT & Data-Breach Monitoring Platform | *FastAPI, PostgreSQL, Celery, Wazuh, React*

- Spearheaded backend architecture for an automated OSINT platform using FastAPI, Celery, and PostgreSQL to aggregate, index, and monitor exposed digital assets.
- Constructed security dashboards with Supabase Row-Level Security (RLS) to enforce strict role-based data partitioning for threat intelligence records.

Devfolio – Secure Authentication Platform | *FastAPI, PostgreSQL, Argon2, OAuth 2.0*

- Implemented cryptographic credential storage using memory-hard Argon2 password hashing and integrated third-party GitHub OAuth 2.0 authentication.
- Hardened relational PostgreSQL database schemas with parameterized queries and strict foreign key constraints, eliminating common web application security risks.

Pulse-Chat – Secure Real-Time Messaging Application | *React, TypeScript, FastAPI, WebSockets, JWT*

- Engineered a full-duplex real-time messaging engine with JWT-authenticated WebSocket handshakes, sustaining 100+ concurrent connections in testing.

- Designed multi-room communication architecture with connection lifecycle monitoring and input sanitization to safeguard channels against socket flooding.

Education

WSB University Kraków, Poland
B.Sc. in Cybersecurity Sept 2026 – July 2030
Université Dr. Moulay Tahar Saïda, Algeria
B.Sc. in Computer Science – Ranked 3rd in Cohort of 120+ Sept 2023 – June 2026
Relevant Coursework: Web Security, Computer Networks, Operating Systems, Linux Systems, Cryptography, Data Structures & Algorithms.

Certifications, Languages & Achievements

Certifications: Google Cybersecurity Professional Certificate (2024); Huawei HCIA-Datacom (2025); Simplilearn CCNA Network Associate (2025); NVIDIA Fundamentals of Deep Learning (2025)
Achievements: 2nd Place Winner – DevRush 1.0 National Programming Competition (2025); National Finalist – FORSA Algérie Télécom Hackathon (2025)
Languages: Arabic (Native), English (Professional – IELTS Academic 6.5, EFSET C1), French (Advanced – TCF C1)